Coursework Solution CID: 02518058

1(a)

Use integrating factor e^t

$$\begin{aligned} d(e^t X(t)) &= e^t dX(t) + X(t) de^t \\ &= e^t[(\mu - X(t))dt + \sigma dW(t)] + X(t)e^t dt \\ &= \mu e^t dt + \sigma e^t dW(t). \end{aligned}$$

Integrate both sides

$$\begin{aligned} e^t X(t) - e^0 X(0) &= \mu \int_0^t e^s ds + \sigma \int_0^t e^s dW(s) \\ &= x_0 + \mu(e^t - 1) + \sigma \int_0^t e^s dW(s). \end{aligned}$$

Thus

$$X(t) = x_0 e^{-t} + \mu(1 - e^{-t}) + \sigma \int_0^t e^{-(t-s)} dW(s)$$

1(b)

We have

$$X(t) = x_0 e^{-t} + \mu(1 - e^{-t}) + \sigma \int_0^t e^{-(t-s)} dW(s)$$

Since $\langle dW(s) \rangle = 0$, taking expectations give

$$\begin{aligned} \langle X(t) \rangle &= x_0 e^{-t} + \mu(1 - e^{-t}) \\ &= \mu + (x_0 - \mu)e^{-t} \end{aligned}$$

For variance, use $dW(t)^2 = dt$

$$X(t) - \langle X(t) \rangle = \sigma \int_0^t e^{-(t-s)} dW(s)$$

$$\langle [X(t) - \langle X(t) \rangle]^2 \rangle = \sigma^2 \left\langle \left[\int_0^t e^{-(t-s)} dW(s) \right]^2 \right\rangle$$
$$= \sigma^2 \int_0^t e^{-2(t-s)} ds = \frac{\sigma^2}{2}(1 - e^{-2t})$$

1(c)

Generator of the diffusion process is defined as

$$\mathbb{L} := A(x)\frac{\partial}{\partial x} + \frac{1}{2}\Sigma(x)\frac{\partial^2}{\partial x^2}$$

For $f \in C_b^2(\mathbb{R})$

$$\mathbb{L}f = (\mu - x)f'(x) + \frac{\sigma^2}{2}f''(x)$$

The L^2-adjoint of the generator $\mathbb{L}^*$ is

$$\mathbb{L}^* p = -\frac{\partial}{\partial x}[(\mu - x)p(x,t)] + \frac{\sigma^2}{2}\frac{\partial^2}{\partial x^2}p(x,t)$$

Hence, the forward Fokker-Planck equation for density $p(x,t)$ is

$$\frac{\partial p}{\partial t} = \mathbb{L}^* p = -\frac{\partial}{\partial x}[(\mu - x)p] + \frac{\sigma^2}{2}\frac{\partial^2 p}{\partial x^2},$$

with initial condition

$$p(x, 0 \mid x_0) = \delta(x - x_0)$$

1(d)

We can write the forward FP equation in conservative form as

$$\frac{\partial p(x,t)}{\partial t} = -\frac{\partial J(x,t)}{\partial x},$$

where $J(x,t)$ is the probability flux, defined as

$$J(x,t) = (\mu - x)p(x,t) - \frac{\sigma^2}{2}\frac{\partial p(x,t)}{\partial x}$$

Steady state means probabiliy flux is zero $J(x,t) = 0$, hence

$$(\mu - x)p(x) - \frac{\sigma^2}{2}p'(x) = 0$$
$$\frac{p'(x)}{p(x)} = \frac{2}{\sigma^2}(\mu - x)$$

Integrate both sides, we obtain

$$\ln p(x) = \frac{2\mu}{\sigma^2}x - \frac{x^2}{\sigma^2} + C$$

We know

$$x^2 - 2\mu x = (x-\mu)^2 - \mu^2$$
$$-\frac{x^2 - 2\mu x}{\sigma^2} = -\frac{(x-\mu)^2}{\sigma^2} + \frac{\mu^2}{\sigma^2}$$

So

$$p(x) = Ce^{\mu^2/\sigma^2}\exp\left(-\frac{(x-\mu)^2}{\sigma^2}\right)$$
$$p(x) = C\exp\left(-\frac{(x-\mu)^2}{\sigma^2}\right)$$

We can normalise it with a normalisation constant

$$p(x) = \mathcal{N}\exp\left(-\frac{(x-\mu)^2}{\sigma^2}\right)$$

Compare to standard gaussian

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

We have mean μ and variance $\sigma^2/2$.

They are are consistent with the $t \to \infty$ limits from part (b):

$$\mathbb{E}[X(t)] = \mu + (x_0 - \mu)e^{-t} \to \mu$$

$$\mathrm{Var}[X(t)] = \frac{\sigma^2}{2}(1 - e^{-2t}) \to \frac{\sigma^2}{2}$$

2(a)

Using Ito's formula,

$$df(X) = \left[A(X)f'(X) + \frac{1}{2}B(X)^2 f''(X)\right] dt + B(X)f'(X)dW(t)$$

With

$$A(x) = \frac{1}{4} - x, B(x) = \sqrt{x}$$

Let $f(x) = 2\sqrt{x}$. Then

$$f'(x) = \frac{1}{\sqrt{x}}, \quad f''(x) = -\frac{1}{2}x^{-3/2}$$

Calcualte each terms

$$B(X)f'(X)dW(t) = \sqrt{X(t)} \cdot \frac{1}{\sqrt{X(t)}} dW(t) = dW(t)$$

$$A(X)f'(X) = \frac{\frac{1}{4} - X(t)}{\sqrt{X(t)}} = \frac{1}{4\sqrt{X(t)}} - \sqrt{X(t)}$$

$$\frac{1}{2}B(X)^2 f''(X) = \frac{1}{2}X(t)\left(-\frac{1}{2}X(t)^{-3/2}\right) = -\frac{1}{4\sqrt{X(t)}}$$

Combine:

$$dY(t) = -\sqrt{X(t)}dt + dW(t)$$
$$dY(t) = -\frac{1}{2}Y(t)dt + dW(t)$$

Use integrating factor $e^{t/2}$

$$\begin{aligned} d(e^{t/2}Y(t)) &= e^{t/2}dY(t) + Y(t)de^{t/2} \\ &= e^{t/2}[-\frac{1}{2}Y(t)dt + dW(t)] + \frac{1}{2}Y(t)e^{t/2}dt \\ &= e^{t/2}dW(t) \end{aligned}$$

Integrate both sides

$$e^{t/2}Y(t) = Y_0 + \int_0^t e^{s/2}dW(s)$$

Thus

$$Y(t) = Y_0e^{-t/2} + \int_0^t e^{-(t-s)/2}dW(s), \quad Y_0 = 2\sqrt{x}.$$

Transforming back

$$X(t) = \frac{1}{4}Y(t)^2 = \frac{1}{4}\left(2\sqrt{x}e^{-t/2} + \int_0^t e^{-(t-s)/2}dW(s)\right)^2$$

2(b)

The general Stratonovich version of SDE has the form

$$dX(t) = A(X)dt + B(X) \circ dW(t)$$

In this question, we have

$$A(X) = 0, \quad B(X) = f(X(t))$$

Given the equivalent Itô SDE of the form

$$dX = \left[A(X) + \alpha B(X)\frac{\partial B(X)}{\partial x}\right]dt + B(X)dW(t)$$

With $\alpha = 0.5$ for Stratonovich, we get

$$dX(t) = \frac{1}{2}f(X(t))f'(X(t))dt + f(X(t))dW(t)$$

Using change of variables $Y(t) = F(X(t))$ and the hint $F(X(t)) = \int^{X(t)} \frac{1}{f(y)} dy$

$$F'(X) = \frac{1}{f(X)}, \quad F''(X) = -\frac{f'(X)}{f(X)^2}.$$

Do taylor expansion

$$\begin{aligned} df(X) &= f(X + dX) - f(X) \\ &= f'(X)dX + \frac{1}{2}f''(X)dX^2 + \dots \end{aligned}$$

So

$$dY(t) = F'(X(t))dX(t) + \frac{1}{2}F''(X(t))(dX(t))^2.$$

First term

$$F'(X)dX = \frac{1}{f(X)}\left[\frac{1}{2}f(X)f'(X)dt + f(X)dW\right] = \frac{1}{2}f'(X)dt + dW.$$

Second term

$$(dX)^2 = f(X)^2 dW^2 = f(X)^2 dt$$

$$\frac{1}{2}F''(X)(dX)^2 = \frac{1}{2}\left(-\frac{f'(X)}{f(X)^2}\right) f(X)^2 dt = -\frac{1}{2}f'(X)dt.$$

Combine and we get

$$\begin{gathered} dY(t) = \left(\frac{1}{2}f'(X(t)) - \frac{1}{2}f'(X(t))\right) dt + dW(t) = dW(t) \\ Y(0) = F(x_0). \end{gathered}$$

Integrating and we get

$$Y(t) = F(x_0) + W(t).$$

Hence

$$X(t) = F^{-1}(Y(t)) = F^{-1}(F(x_0) + W(t)), \quad F(x) = \int^x \frac{1}{f(y)} dy.$$

2(c)

If

$$dX(t) = f(X(t)) \circ dW(t)$$

then the equivalent Itô SDE for Stratonovich is

$$dX(t) = \frac{1}{2} f(X(t)) f'(X(t)) dt + f(X(t)) dW(t)$$

Take

$$f(x) = \sqrt{1 - x^2}$$

Get derivatives

$$f'(x) = -\frac{x}{\sqrt{1 - x^2}}$$

$$\frac{1}{2} f(x) f'(x) = \frac{1}{2} \sqrt{1 - x^2} \left(-\frac{x}{\sqrt{1 - x^2}} \right) = -\frac{1}{2} x$$

So the corresponding Ito SDE is

$$dX(t) = -\frac{1}{2} X(t) dt + \sqrt{1 - X(t)^2} dW(t)$$

which is exactly the given Ito SDE. Therefore, we can write

$$dX(t) = \sqrt{1 - X(t)^2} \circ dW(t)$$

From part (b), the solution of this form is

$$X(t) = F^{-1}(F(x_0) + W(t))$$

where

$$F(x) = \int^x \frac{1}{f(y)} dy = \int^x \frac{1}{\sqrt{1 - y^2}} dy = \arcsin(x)$$

So let

$$F(x) = \arcsin(x), \quad F^{-1}(u) = \sin(u)$$

Hence

$$X(t) = F^{-1}(F(x_0) + W(t)) = \sin(W(t) + \arcsin(x_0))$$

3(a)

Drift vector:

$$\mathbf{A}(x, y, z) = \begin{pmatrix} y \\ -x + z \\ -z - y \end{pmatrix}$$

There is one Brownian motion with coefficient vector

$$B(x, y, z) = \begin{pmatrix} 0 \\ 0 \\ \sqrt{2} \end{pmatrix}$$

so

$$BB^\top = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

Diffusion matrix $(\frac{1}{2}(\sqrt{2})^2 = 1)$

$$\mathbf{\Sigma} = \frac{1}{2} BB^\top = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

The generator $\mathbb{L}$ acting on a smooth test function $f(x, y, z)$ is

$$\begin{aligned}\mathbb{L}f &= \mathbf{A}\cdot\nabla f + \frac{1}{2}\mathrm{Tr}(BB^\top\nabla^2 f)\\ &= \sum_{j=1}^{d} A_j\frac{\partial f}{\partial y_j} + \frac{1}{2}\sum_{i,j=1}^{d}(BB^\top)_{ij}\frac{\partial^2 f}{\partial y_i\partial y_j}\\ &= y\frac{\partial f}{\partial x} + (z-x)\frac{\partial f}{\partial y} - (y+z)\frac{\partial f}{\partial z} + \frac{\partial^2 f}{\partial z^2}\end{aligned}$$

The L^2-adjoint $\mathbb{L}^*$is

$$\begin{aligned}\mathbb{L}^* f &= -\nabla\cdot(\mathbf{A}f) + \frac{1}{2}\mathrm{Tr}(BB^\top\nabla^2 f))\\ &= -\sum_{j=1}^{d}\frac{\partial(A_j f)}{\partial y_j} + \frac{1}{2}\sum_{i,j=1}^{d}(BB^\top)_{ij}\frac{\partial^2 f}{\partial y_i\partial y_j}\\ &= -\frac{\partial}{\partial x}(yf) - \frac{\partial}{\partial y}((z-x)f) - \frac{\partial}{\partial z}((-y-z)f) + \frac{\partial^2 f}{\partial z^2}\end{aligned}$$

The forward (Fokker-Planck) Kolmogorov equation for joint probability density $p(x,y,z,t)$ is

$$\frac{\partial p}{\partial t} = \mathbb{L}^* p = -y\frac{\partial p}{\partial x} + (x-z)\frac{\partial p}{\partial y} + \frac{\partial}{\partial z}((y+z)p) + \frac{\partial^2 p}{\partial z^2}$$

With initial conditions

$$X(0) = x_0, Y(0) = y_0, Z(0) \sim \mathcal{N}(0,1)$$

And initial density

$$p(x,y,z,0) = \delta(x-x_0)\delta(y-y_0)\frac{1}{\sqrt{2\pi}}e^{-z^2/2}$$

3(b)

Use the integrating factor e^t

$$\begin{aligned}d(e^t Z(t)) &= e^t dZ(t) + e^t Z(t)dt = e^t(dZ + Zdt)\\ &= -e^t Y(t)dt + \sqrt{2}e^t dW(t)\end{aligned}$$

Integrate

$$e^t Z(t) = Z(0) - \int_0^t e^s Y(s) ds + \sqrt{2} \int_0^t e^s dW(s)$$

So

$$Z(t) = e^{-t} Z(0) - \int_0^t e^{-(t-s)} Y(s) ds + \sqrt{2} \int_0^t e^{-(t-s)} dW(s)$$

Let

$$F(t) = e^{-t} Z(0) + \sqrt{2} \int_0^t e^{-(t-s)} dW(s)$$

then

$$Z(t) = F(t) - \int_0^t e^{-(t-s)} Y(s) ds$$

Sub into dY/dt

$$\frac{dY(t)}{dt} = -X(t) + Z(t) = -X(t) - \int_0^t e^{-(t-s)} Y(s) ds + F(t)$$

Also from $dX = Y dt$

$$\frac{dX(t)}{dt} = Y(t)$$

Assume $t \geq s$, then

$$F(t) = e^{-t} Z(0) + \sqrt{2} \int_0^t e^{-(t-u)} dW(u)$$
$$F(s) = e^{-s} Z(0) + \sqrt{2} \int_0^s e^{-(s-v)} dW(v)$$

$$\begin{aligned}\mathbb{E}[F(t)F(s)] = e^{-t-s}\mathbb{E}[Z(0)^2] \\ &+ \sqrt{2}e^{-t}\mathbb{E}\left[Z(0)\int_0^s e^{-(s-v)}dW(v)\right] \\ &+ \sqrt{2}e^{-s}\mathbb{E}\left[Z(0)\int_0^t e^{-(t-u)}dW(u)\right] \\ &+ 2\mathbb{E}\left[\int_0^t e^{-(t-u)}dW(u)\int_0^s e^{-(s-v)}dW(v)\right]\end{aligned}$$

Because $W(t)$ is a Brownian motion independent of $Z(0)$ and $\mathbb{E}[Z(0)] = 0$, the second and third term vanish.

For the first term:

$$\mathbb{E}[Z(0)^2] = \mathrm{Var}(Z(0)) + (\mathbb{E}[Z(0)])^2 = 1 + 0 = 1$$

so

$$e^{-t-s}\mathbb{E}[Z(0)^2] = e^{-t-s}$$

For the forth term, since we assume $t \geq s$, we can decompose

$$\int_0^t e^{-(t-u)}dW(u) = \int_0^s e^{-(t-u)}dW(u) + \int_s^t e^{-(t-u)}dW(u)$$

The brownian increments from s to t is independent of the increments from 0 to s, so the expectations of their product is zero, so we just need to handle the interval from 0 to s only or take $\min(t, s) = s$, so

$$\begin{aligned}&\mathbb{E}\left[\int_0^t e^{-(t-u)}dW(u)\int_0^s e^{-(s-v)}dW(v)\right] \\ &= \int_0^s e^{-(t-u)}e^{-(s-u)}du = \int_0^s e^{-(t+s-2u)}du \\ &= e^{-(t+s)}\int_0^s e^{2u}du = e^{-(t+s)}\cdot\frac{e^{2s}-1}{2}\end{aligned}$$

Hence,

$$\begin{aligned}\mathbb{E}[F(t)F(s)] &= e^{-t-s} + 2\cdot e^{-(t+s)}\cdot\frac{e^{2s}-1}{2} \\ &= e^{-t-s} + e^{-(t-s)} - e^{-(t+s)} = e^{-(t-s)}\end{aligned}$$

For $s > t$, by the same argument, we obtain $e^{-(s-t)}$, hence

$$\mathbb{E}[F(t)F(s)] = e^{-|t-s|}$$

3(c)

A continuous stochastic process is said to satisfy the Markov property if

$$p(x_n, t_n | x_1, \ldots, x_{n-1}; t_1, \ldots, t_{n-1}) = p(x_n, t_n | x_{n-1}, t_{n-1})$$

So in a Markov process, the change at time t must only depends on the current state $(X(t), Y(t))$ and current noise increment.

However, $(X(t), Y(t))$ is not a Markov process because looking at the equation for Y, the drift term here

$$\int_0^t e^{-(t-s)} Y(s) ds$$

contain all memory about the past $0 \leq s \leq t$, so it does not satisfy the Markov property. The exponential term $e^{-(t-s)}$ put weights on the past values that decay exponentially.

Also, the driving term $F(t)$ is not white noise, it is colored noised with an autocorrelation of $e^{-|t-s|}$. This mens that the force at time t is correlated with the force at previous time.

4(a)

Given the OU process in vector form

$$d\mathbf{X}(t) = \mathbf{A}\mathbf{X}(t)dt + \mathbf{B}d\mathbf{W}(t)$$

Use integrating factor $e^{-\mathbf{A}t}$

$$\begin{aligned} d(e^{-\mathbf{A}t}\mathbf{X}(t)) &= -\mathbf{A}e^{-\mathbf{A}t}\mathbf{X}(t)dt + e^{-\mathbf{A}t}d\mathbf{X}(t) \\ &= -\mathbf{A}e^{-\mathbf{A}t}\mathbf{X}(t)dt + e^{-\mathbf{A}t}(\mathbf{A}\mathbf{X}(t)dt + \mathbf{B}d\mathbf{W}(t)) \\ &= e^{-\mathbf{A}t}\mathbf{B}d\mathbf{W}(t) \end{aligned}$$

Integrate

$$e^{-\mathbf{A}t}\mathbf{X}(t) = \mathbf{X}(0) + \int_0^t e^{-\mathbf{A}t'}\mathbf{B}d\mathbf{W}(t')$$
$$\mathbf{X}(t) = e^{\mathbf{A}t}\bar{\mathbf{x}} + \int_0^t e^{\mathbf{A}(t-t')}\mathbf{B}d\mathbf{W}(t')$$

Where $\mathbf{X}(0) = \bar{\mathbf{x}}$

4(b)

Taking expectation on both side

$$\langle \mathbf{X}(t)\rangle = e^{\mathbf{A}t}\bar{\mathbf{x}} + \int_0^t e^{\mathbf{A}(t-t')}\mathbf{B}\langle d\mathbf{W}(t')\rangle$$

Since $\langle d\mathbf{W}(t')\rangle = 0$, we have

$$\langle \mathbf{X}(t)\rangle = e^{\mathbf{A}t}\bar{\mathbf{x}}$$

The centered process is

$$\mathbf{X}(t) - \langle \mathbf{X}(t)\rangle = \int_0^t e^{\mathbf{A}(t-t')}\mathbf{B}d\mathbf{W}(t')$$
$$\mathbf{X}(s) - \langle \mathbf{X}(s)\rangle = \int_0^s e^{\mathbf{A}(s-u)}\mathbf{B}d\mathbf{W}(u)$$
$$(\mathbf{X}(s) - \langle \mathbf{X}(s)\rangle)^\top = \int_0^s d\mathbf{W}(u)^\top\mathbf{B}^\top e^{\mathbf{A}^\top(s-u)}$$

Hence the correlation function

$$\mathbf{C}(t,s) = \left\langle \left(\int_0^t e^{\mathbf{A}(t-t')}\mathbf{B}d\mathbf{W}(t')\right)\left(\int_0^s d\mathbf{W}(u)^\top\mathbf{B}^\top e^{\mathbf{A}^\top(s-u)}\right)\right\rangle$$

For a d-dimensional standard Brownian motion, we know that

$$\mathbb{E}[W_i(t)W_j(s)] = \delta_{i,j}\min\{t,s\}$$
$$\mathbb{E}\left[\int_0^t F(t')d\mathbf{W}(t')\int_0^s d\mathbf{W}(u)^\top G(u)^\top\right] = \int_0^{\min(t,s)} F(u)G(u)^\top du$$

because the nonoverlapping region simply independent of overlapping region, so the expectations of their product is zero, as explain in question 3b answer, so just take 0 to $\min(t, s)$

$$\mathbf{C}(t, s) = \int_0^{\min(t,s)} e^{\mathbf{A}(t-t')}\mathbf{B}\mathbf{B}^\top e^{\mathbf{A}^\top(s-t')}dt'$$

4(c)

From (b)

$$\Sigma(t) = \mathbf{C}(t, t) = \int_0^t e^{\mathbf{A}(t-t')}\mathbf{B}\mathbf{B}^\top e^{\mathbf{A}^\top(t-t')}dt'$$

Differentiate with respect $\Sigma(t)$ to t

$$\frac{d\Sigma(t)}{dt} = \frac{d}{dt}\int_0^t e^{\mathbf{A}(t-t')}\mathbf{B}\mathbf{B}^\top e^{\mathbf{A}^\top(t-t')}dt'$$

Since there is no stochastic term, we use Leibniz rule

$$\frac{d}{dt}\left(\int_0^{b(t)} f(t, u)du\right) = f(t, b(t))b'(t) + \int_0^{b(t)} \frac{\partial}{\partial t}f(t, u)du$$

Evaluate upper limit term, sub $t' = t$,

$$e^{\mathbf{A}(t-t)}\mathbf{B}\mathbf{B}^\top e^{\mathbf{A}^\top(t-t)} = \mathbf{B}\mathbf{B}^\top$$

Evaluate derivative of the integrand

$$\begin{aligned}&\frac{\partial}{\partial t}\left(e^{\mathbf{A}(t-t')}\mathbf{B}\mathbf{B}^\top e^{\mathbf{A}^\top(t-t')}\right)\\ &= \mathbf{A}e^{\mathbf{A}(t-t')}\mathbf{B}\mathbf{B}^\top e^{\mathbf{A}^\top(t-t')} + e^{\mathbf{A}(t-t')}\mathbf{B}\mathbf{B}^\top e^{\mathbf{A}^\top(t-t')}\mathbf{A}^\top\end{aligned}$$

Now

$$\frac{d\Sigma(t)}{dt} = \mathbf{B}\mathbf{B}^\top + \int_0^t \left[\mathbf{A}e^{\mathbf{A}(t-t')}\mathbf{B}\mathbf{B}^\top e^{\mathbf{A}^\top(t-t')} + e^{\mathbf{A}(t-t')}\mathbf{B}\mathbf{B}^\top e^{\mathbf{A}^\top(t-t')}\mathbf{A}^\top\right]dt'$$

$$\int_0^t \mathbf{A}e^{\mathbf{A}(t-t')}\mathbf{B}\mathbf{B}^\top e^{\mathbf{A}^\top(t-t')}dt' = \mathbf{A}\Sigma(t)$$

$$\int_0^t e^{\mathbf{A}(t-t')}\mathbf{B}\mathbf{B}^\top e^{\mathbf{A}^\top(t-t')}\mathbf{A}^\top dt' = \Sigma(t)\mathbf{A}^\top$$

Therefore,

$$\frac{d\Sigma(t)}{dt} = \mathbf{A}\Sigma(t) + \Sigma(t)\mathbf{A}^\top + \mathbf{B}\mathbf{B}^\top$$

To show $\Sigma(t) \to \Sigma_0$, let $\tau = t - t'$. Then $d\tau = -dt'$

- When $t' = 0$, $\tau = t$
- When $t' = t$, $\tau = 0$

Substitute these into the integral, we get

$$\Sigma(t) = \int_0^t e^{\mathbf{A}\tau}\mathbf{B}\mathbf{B}^\top e^{\mathbf{A}^\top\tau} d\tau$$

Define the limit as $t \to \infty$

$$\Sigma_0 = \lim_{t\to\infty} \Sigma(t) = \int_0^\infty e^{\mathbf{A}\tau}\mathbf{B}\mathbf{B}^\top e^{\mathbf{A}^\top\tau} d\tau$$

Now do

$$\mathbf{A}\Sigma_0 + \Sigma_0\mathbf{A}^\top$$
$$= \mathbf{A}\left(\int_0^\infty e^{\mathbf{A}\tau}\mathbf{B}\mathbf{B}^\top e^{\mathbf{A}^\top\tau} d\tau\right) + \left(\int_0^\infty e^{\mathbf{A}\tau}\mathbf{B}\mathbf{B}^\top e^{\mathbf{A}^\top\tau} d\tau\right)\mathbf{A}^\top$$

Since $\mathbf{A}$ is constant, we can move inside the integral

$$= \int_0^\infty \left(\mathbf{A}e^{\mathbf{A}\tau}\mathbf{B}\mathbf{B}^\top e^{\mathbf{A}^\top\tau} + e^{\mathbf{A}\tau}\mathbf{B}\mathbf{B}^\top e^{\mathbf{A}^\top\tau}\mathbf{A}^\top\right) d\tau$$

The integrand is the derivative of the term inside the integral by product rule

$$\frac{d}{d\tau}\left(e^{\mathbf{A}\tau}\mathbf{B}\mathbf{B}^\top e^{\mathbf{A}^\top\tau}\right) = \mathbf{A}e^{\mathbf{A}\tau}\mathbf{B}\mathbf{B}^\top e^{\mathbf{A}^\top\tau} + e^{\mathbf{A}\tau}\mathbf{B}\mathbf{B}^\top e^{\mathbf{A}^\top\tau}\mathbf{A}^\top$$

Now evaluate the integral

$$\mathbf{A}\Sigma_0 + \Sigma_0\mathbf{A}^\top = \int_0^\infty \frac{d}{d\tau}\left(e^{\mathbf{A}\tau}\mathbf{B}\mathbf{B}^\top e^{\mathbf{A}^\top\tau}\right) d\tau = \left[e^{\mathbf{A}\tau}\mathbf{B}\mathbf{B}^\top e^{\mathbf{A}^\top\tau}\right]_0^\infty$$

For the upper limit, since the eigenvalues of $\mathbf{A}$ have negative real parts, the matrix exponential $e^{\mathbf{A}\tau}$ must decays to zero as $\tau \to \infty$, so $e^{\mathbf{A}\tau} \to 0$. Term vanish.

For the lower limit, $\mathbf{I}$, so

$$\mathbf{I}\mathbf{B}\mathbf{B}^\top\mathbf{I} = \mathbf{B}\mathbf{B}^\top$$

Therefore,

$$\begin{aligned}\mathbf{A}\Sigma_0 + \Sigma_0\mathbf{A}^\top &= 0 - \mathbf{B}\mathbf{B}^\top \\ \mathbf{A}\Sigma_0 + \Sigma_0\mathbf{A}^\top + \mathbf{B}\mathbf{B}^\top &= 0\end{aligned}$$

4(d)

Rewrite as

$$\frac{d\mathbf{X}(t)}{dt} = \mathbf{A}\mathbf{X}(t) + \mathbf{B}\boldsymbol{\xi}(t).$$

Define the Fourier transform of a function $f(t)$ as

$$\tilde{f}(\omega) = \int_{-\infty}^{\infty} f(t)e^{i\omega t}dt$$

Take Fourier transform of the SDE

$$-i\omega\tilde{\mathbf{X}}(\omega) = \mathbf{A}\tilde{\mathbf{X}}(\omega) + \mathbf{B}\tilde{\boldsymbol{\xi}}(\omega).$$

Rearrange

$$(-i\omega\mathbf{I} - \mathbf{A})\tilde{\mathbf{X}}(\omega) = \mathbf{B}\tilde{\boldsymbol{\xi}}(\omega)$$

Substitute the transfer matrix

$$\mathbf{\Phi}_{ij}(\omega) = -i\omega\delta_{i,j} - A_{ij}$$

So

$$\mathbf{\Phi}(\omega)\tilde{\mathbf{X}}(\omega) = \mathbf{B}\tilde{\boldsymbol{\xi}}(\omega)$$

Invert the matrix

$$\tilde{\mathbf{X}}(\omega) = \mathbf{\Phi}^{-1}(\omega)\mathbf{B}\tilde{\boldsymbol{\xi}}(\omega)$$

In component form

$$\tilde{X}_i(\omega) = \sum_{j=1}^{d}\sum_{k=1}^{d} \Phi_{ij}^{-1}(\omega) B_{jk}\tilde{\xi}_k(\omega).$$

Consider the covariance of two frequency components of $X(t)$

$$\langle \tilde{X}_i(\omega)\tilde{X}_i(\omega')\rangle = \left\langle \left(\sum_{j,k} \Phi_{ij}^{-1}(\omega)B_{jk}\tilde{\xi}_k(\omega)\right)\left(\sum_{j',k'} \Phi_{ij'}^{-1}(\omega')B_{j'k'}\tilde{\xi}_{k'}(\omega')\right)\right\rangle$$

Since Φ^{-1} and B are deterministic, we move them out of the expectation,

$$\langle \tilde{X}_i(\omega)\tilde{X}_i(\omega')\rangle = \sum_{j,j'}\sum_{k,k'} \Phi_{ij}^{-1}(\omega)B_{jk}\Phi_{ij'}^{-1}(\omega')B_{j'k'}\langle\tilde{\xi}_k(\omega)\tilde{\xi}_{k'}(\omega')\rangle$$

Calculate noise correlation in the frequency domain using the time-domain properties provided: $\langle \xi_k(t)\xi_{k'}(t')\rangle = \delta_{kk'}\delta(t-t')$

$$\begin{aligned}
\langle\tilde{\xi}_k(\omega)\tilde{\xi}_{k'}(\omega')\rangle &= \int_{-\infty}^{\infty}dt\int_{-\infty}^{\infty}dt' e^{i\omega t}e^{i\omega' t'}\langle\xi_k(t)\xi_{k'}(t')\rangle \\
&= \delta_{kk'}\int_{-\infty}^{\infty}dt\int_{-\infty}^{\infty}dt' e^{i\omega t}e^{i\omega' t'}\delta(t-t') \\
&= \delta_{kk'}\int_{-\infty}^{\infty}dt\, e^{i(\omega+\omega')t} \\
&= \delta_{kk'}2\pi\delta(\omega+\omega').
\end{aligned}$$

Substitute back and we get

$$\langle \tilde{X}_i(\omega)\tilde{X}_i(\omega')\rangle = \sum_{j,j'} \Phi_{ij}^{-1}(\omega)\Phi_{ij'}^{-1}(\omega')\left(\sum_k B_{jk}B_{j'k}\right)2\pi\delta(\omega+\omega')$$

Noted that the term $\delta_{kk'}$ forces $k=k'$, s o we get single sum

Since the diffusion matrix term is

$$\sum_k B_{jk}B_{j'k} = (\mathbf{B}\mathbf{B}^\top)_{jj'} = D_{jj'}$$

We now have

$$\langle \tilde{X}_i(\omega)\tilde{X}_i(\omega')\rangle = \left[\sum_{j,j'} \Phi_{ij}^{-1}(\omega)\Phi_{ij'}^{-1}(\omega')D_{jj'}\right] 2\pi\delta(\omega+\omega').$$

The Wiener-Khinchin theorem states:

$$\langle \tilde{X}_i(\omega)\tilde{X}_i(\omega')\rangle = 2\pi S_i(\omega)\delta(\omega+\omega')$$

Comparing terms, and noting that the Dirac delta implies that the expression is non-zero only when $\omega' = -\omega$, hence we get

$$S_i(\omega) = \sum_{j=1}^{d}\sum_{j'=1}^{d} \Phi_{ij}^{-1}(\omega)\Phi_{ij'}^{-1}(-\omega)D_{jj'}$$

5(a)

Drfit and diffusion coefficient are

$$A(x) = -x^3 + x, \quad \Sigma(x) = (\sqrt{2})^2 = 2$$

Generator is

$$\mathbb{L} = A(x)\partial_x + \frac{1}{2}\Sigma(x)\partial_x^2 = (-x^3+x)\partial_x + \partial_x^2.$$

Backward Kolmogorov equation for $u(x,t) = \mathbb{E}(f(X(t) \mid X(0) = x)$ is

$$\frac{\partial u}{\partial t} = \mathbb{L}u = (-x^3+x)\frac{\partial u}{\partial x} + \frac{\partial^2 u}{\partial x^2}$$

Forward Kolmogorov equation for density $p(x,t)$ is

$$\frac{\partial p}{\partial t} = \mathbb{L}^* p = -\frac{\partial}{\partial x}[(-x^3+x)p(x,t)] + \frac{\partial^2 p}{\partial x^2}, \quad p(x,0) = p_0(x).$$

5(b)

$$m_n(t) = \mathbb{E}[X(t)^n] = \int_{\mathbb{R}} x^n p(x,t)dx$$

$$\frac{d}{dt}m_n(t) = \int_{\mathbb{R}} x^n \frac{\partial p}{\partial t}(x,t)dx$$

$$\frac{d}{dt}m_n(t) = -\int_{\mathbb{R}} x^n \frac{\partial}{\partial x}[(-x^3+x)p]dx + \int_{\mathbb{R}} x^n \frac{\partial^2 p}{\partial x^2}dx = I_1 + I_2$$

Integrate by parts for drift term I_1

$$u = x^n, \quad dv = \partial_x[(-x^3+x)p]dx$$

$$du = nx^{n-1}dx, \quad v = (-x^3+x)p$$

$$I_1 = -[x^n(-x^3+x)p]_{-\infty}^{\infty} + \int (-x^3+x)pnx^{n-1}dx$$

assume p vanish sufficiently fast at boundary, so boundary term vanishes

$$-[x^n(-x^3+x)p]_{-\infty}^{\infty} = 0$$

so

$$\begin{aligned} I_1 &= n\int_{\mathbb{R}} x^{n-1}(-x^3+x)pdx \\ &= n\mathbb{E}[X^{n-1}(-X^3+X)] \\ &= n\mathbb{E}[-X^{n+2}+X^n] \\ &= n(-m_{n+2}(t)+m_n(t)) \end{aligned}$$

Integrate by parts twice for diffusion term I_2

$$u = x^n, \quad dv = p_{xx}dx$$

$$du = nx^{n-1}dx, \quad v = p_x$$

$$I_2 = [x^n p_x]_{-\infty}^{\infty} - \int nx^{n-1}p_x dx$$

assume p and its detivatives decay sufficiently fast as $|x| \to \infty$, so the boundary term vanish

$$[x^n p_x]_{-\infty}^{\infty} = 0$$

so

$$I_2 = -n \int x^{n-1} p_x dx$$

$$u = x^{n-1}, \quad dv = p_x dx$$

$$du = (n-1)x^{n-2}dx, \quad v = p$$

$$I_2 = -n[x^{n-1}p]_{-\infty}^{\infty} + n \int (n-1)x^{n-2} p dx$$

Again the boundary term vanishes, so

$$\begin{aligned} I_2 &= n(n-1) \int x^{n-2} p(x,t) dx \\ &= n(n-1) m_{n-2}(t) \end{aligned}$$

Combine

$$\frac{d}{dt} m_n(t) = I_1 + I_2 = n(-m_{n+2}(t) + m_n(t)) + n(n-1)m_{n-2}(t)$$

For $n = 0$

$$\frac{d}{dt} m_0(t) = 0$$

Since $m_0(t) = \mathbb{E}[X^0] = \int p(x,t)dx = \int p_0(x)dx = 1$, so

$$m_0(t) = m_0(0) = 1$$

For $n = 1$

$$\frac{d}{dt} m_1(t) = m_1(t) - m_3(t)$$

the first moment depends on the third moment

For $n = 2$

$$\begin{aligned} \frac{d}{dt} m_2(t) &= 2(-m_4(t) + m_2(t)) + 2 \cdot 1 \cdot m_0(t) \\ &= 2(m_2(t) - m_4(t)) + 2m_0(t) \end{aligned}$$

Using $m_0(t) = 1$, this becomes

$$\frac{d}{dt}m_2(t) = 2(m_2(t) - m_4(t)) + 2$$

second moment also depends on higher moment term.

5(c)

At stationary, $\partial_t p = 0$, so

$$0 = -\frac{d}{dx}A(x)p(x) + \frac{d^2}{dx^2}p(x)$$

$$0 = \frac{d}{dx}\left(-A(x)p(x) + \frac{d}{dx}p(x)\right)$$

$$\frac{d}{dx}(-A(x)p(x) + p'(x)) = 0$$

$$-A(x)p(x) + p'(x) = C$$

Since $p(x)$ is a probability density on $\mathbb{R}$ that decays (and so does $p'(x)$) as $|x| \to \infty$. Then the LHS goes to 0 as $|x| \to \infty$, so

$$C = 0$$

So

$$\frac{p'(x)}{p(x)} = A(x)$$

$$\frac{d}{dx}(\ln p(x)) = A(x)$$

Integrate:

$$\ln p(x) = \int A(x)dx + C_1$$

$$p(x) = e^{C_1}\exp\left(\int A(x)dx\right)$$

Define

$$V(x) = -\int A(x)dx = -\int(-x^3 + x) = \frac{x^4}{4} - \frac{x^2}{2} + C_2$$

Absorb the constant C_2 into C_1, so we take

$$V(x) = \frac{x^4}{4} - \frac{x^2}{2}$$

Then

$$p(x) = e^{C_1}e^{-V(x)}$$

Define normalisation constant such that $\int p(x)dx = 1$

$$e^{C_1} = Z^{-1}, \quad Z = \int_{\mathbb{R}} e^{-V(x)}dx$$

Hence

$$p^*(x) = Z^{-1}e^{-V(x)}$$

$$p^*(x) = Z^{-1}\exp(-\frac{x^4}{4} + \frac{x^2}{2})$$

5(d)

$$\mathbb{L}^*(h(x)p^*(x)) = 0$$

$$-\frac{d}{dx}[A(x)h(x)p^*(x)] + \frac{d^2}{dx^2}[h(x)p^*(x)] = 0$$

Let $Ahp = A(x)h(x)p^*(x)$ and expand derivatives

First term:

$$-\frac{d}{dx}[h \cdot (Ap)] = -h'(Ap) - h(Ap)'$$

Second term:

$$\frac{d}{dx}[hp] = h'p + hp'$$

$$\begin{aligned}\frac{d^2}{dx^2}[hp] &= \frac{d}{dx}[h'p + hp'] = (h''p + h'p') + (h'p' + hp'') \\ &= h''p + 2h'p' + hp''\end{aligned}$$

Combine:

$$-h'(Ap) - h(Ap)' + h''p + 2h'p' + hp'' = 0$$

Rearrage:

$$h(x)[-(Ap)' + p''] + h'(x)[-Ap + 2p'] + h''(x)p = 0$$

This term $[-(Ap)' + p'']$ is $\mathbb{L}^* p$. Since $p = p^*(x)$ is the stationary density, we have $\mathbb{L}^* p = 0$ so the first $h(x)$ term vanish.

Part (c) also shown that for the stationary density $p = p^*(x)$,

$$-Ap + p' = 0$$

Substitute $p' = Ap$, we get

$$\begin{gathered}h'(x)[-Ap + 2Ap] + h''(x)p = 0 \\ h'(x)[Ap] + h''(x)p = 0\end{gathered}$$

Since p is probability density and is strictyly positive, we divide by p and get

$$A(x)h'(x) + h''(x) = 0$$

Since the generator $\mathbb{L}$ is defined as

$$\mathbb{L} = A(x)\partial_x + \partial_x^2$$

Hence,

$$\mathbb{L}h(x) = A(x)h'(x) + h''(x) = 0$$

Now multiply by $h(x)p^*(x)$ and integrate

$$\int_{\mathbb{R}} h(x)p^*(x)[A(x)h'(x) + h''(x)]dx = 0$$

$$\int_{\mathbb{R}} h(x)h'(x)A(x)p^*(x)dx + \int_{\mathbb{R}} h(x)p^*(x)h''(x)dx = I_1 + I_2 = 0$$

For the first integral, substituite $A(x)p^*(x) = p'^*(x)$

$$I_1 = \int_{\mathbb{R}} h(x)h'(x)p'^*(x)dx$$

For the second integral, do integration by parts

Let $u = h(x)p^*(x) = hp$ and $dv = h''(x)dx = h''dx$.Then

$$du = h'p + hp'$$
$$v = h'$$

$$I_2 = [hph']_{-\infty}^{\infty} - \int_{\mathbb{R}} h'(h'p + hp')dx$$

Since $p(x)$ is a probability density on $\mathbb{R}$ that decays to 0 as $|x| \to \infty$, so boundary term vanish and we get

$$I_2 = -\int_{\mathbb{R}} |h'|^2 p^* dx - \int_{\mathbb{R}} h'hp'^* dx$$

Combine $I_1 + I_2$

$$\int_{\mathbb{R}} hh'p'^* dx - \int_{\mathbb{R}} |h'|^2 p^* dx - \int_{\mathbb{R}} h'hp'^* dx = 0$$

Hence

$$\int_{\mathbb{R}} |h'|^2 p^* dx = 0$$

We know that the stationary density $p^*(x) = Z^{-1}e^{-V(x)}$ is strictly positive for all x

Since $|h'(x)|^2 \geq 0$ and $p^*(x) > 0$, the integral can only be zero if the integrand is zero everywhere. This means that $h(x)$ is a constant:

$$h(x) = C$$

From part (c) we know $p^*(x)$ is normalised and so

$$h(x)p^*(x) = Cp^*(x)$$

Since $h(x)p^*(x)$ is a probability density, it must integrate to 1, and $\int p^*(x)dx = 1$ so $C = 1$

Hence, we prove that the invariant measure is unique and this means that the process $X(t)$ is ergodic.